MIL-M-28787/57A(EC)
25 October 1978
SUPERSEDING
MIL-M-28787/57(NAVY)
8 November 1973

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

SWITCH, ANALOG, ANALOG, KEY CODE TTY

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic analog switch module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Switch, analog
Family	Analog
Key code	TTY
Size	1A
Weight	35 grams maximum
Failure rate	5.30 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage and current	+15 Vdc ±10% at 65 mA maximum -15 Vdc ±10% at 35 mA maximum
Maximum power dissipation	1.65 W
Logic input signal at switch "on" position	2.4 V minimum at 60 µA for each gate, 5.5 V maximum
Logic input signal at switch "off" position	0.8 V maximum at 0.1 µA for each gate, -5.5 V minimum
Analog input signal level	5.0 Vrms maximum -7.0 Vdc minimum +10.0 Vdc maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Analog switch "on" resistance (with V1 only applied)	15Ω maximum
Analog switch "off" resistance	
With 4.7 kΩ load and 5 Vrms, 10 kHz signal	>1.8 MΩ
With 4.7 kΩ load and 17 Vdc across switch	≥10 MΩ
Drain current, switch "on"	30 mAdc maximum 21 mArms maximum
Source or drain leakage current	0.9 µA maximum
Turn "on" time	1.5 µs maximum
Turn "off" time	3.7 µs maximum
Crosstalk	2.83 mV_{p-p}/1 V_{p-p}
Harmonic distortion	0.2% maximum

1.3 Module function. The analog switch module, designated as key code TTY, is a Standard Electronic Modules Program (SEMP) electronic module with 0.100 inch (2.54 mm) grid center contacts. This module contains eight SPST logic controlled, solid state, switches with independent inputs and outputs. Each switch has its own level shifting driver circuit enabling low-level inputs to control the on-off state of the switch. Applying a high-level signal to any of the logic gate inputs will close the switch controlled by that gate. Applying a low-level signal to any of the logic gate inputs will cause the respective switch to open. With the V2 power supply removed and a high-level signal applied to the gate input, the switch is closed. In the closed position, the switch will conduct in either direction.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(All military standards and military specifications shall be ordered from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 5.30 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

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3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be TTY.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/57, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed two years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of ±15 Vdc ±10 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	5
Square wave generator	Item H01	1
Sine wave generator	Item G01	1
Power amplifier	Item A01	1
Megohm bridge	Item R06	1
Distortion analyzer	Item B01	1
AC voltmeter	Item W02	1
Ohmmeter	Item R04	1
Function generator	Item G05	1
Oscilloscope	Item D01	1
DC ammeter	Item X01	3
DC voltmeter	Item V02	1
Pulse generator	Item J01	1
DC voltmeter	Item V05	1

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Resistors	MIL-STD-1665 designator	No. required (maximum per test figure)
235Ω ±1%, 1/2 W		8
4.7 kΩ ±1%, 1/10 W		8
50Ω ±5%, 1/2 W		1
500Ω ±1%, 1/10 W		1
7 kΩ ±1%, 1/10 W		1
1 kΩ ±1%, 1/10 W		1
Capacitor		
35 pF ±10%, 20 Vdc		1
Switches		
SPST		1
DPDT		1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SEMP Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through P in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through P shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

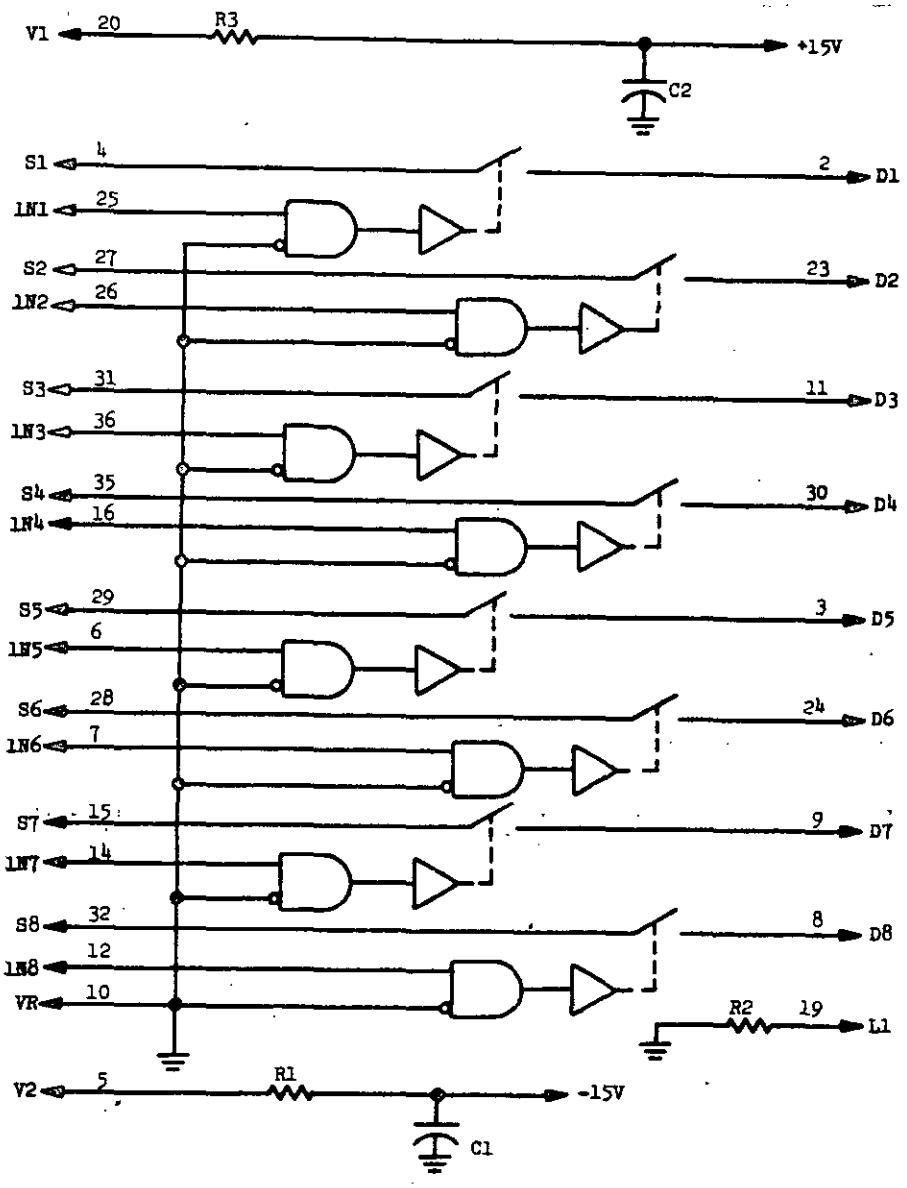
4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 2. Following the vibration tests, the electrical tests of table I which are marked with an (X) in the test type column, and test condition G, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following life test and while still at the maximum class temperature, electrical tests C through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTES:
1. R1 = R3 = 100Ω.
2. R2 = 4.7 kΩ.
3. C1 = C2 = 2.2 µF.
4. The following pins are not used and each shall be isolated from all other pins: 1, 13, 17, 18, 21, 22, 33, 34, 37, 38, 39, and 40.

FIGURE 1. TTY module function diagram, analog switch.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B	Power supply sequence	III	Q	← See table III →								
C1-C16	Analog switch feed thru, switch "off"	IV	X	0.0	12.5	0.0	12.5	0.0	12.5	0.0	12.5	mVrms
D1-D16	Analog switch feed thru, switch "on"	IV	X	4.9	5.1	4.9	5.1	4.9	5.1	4.9	5.1	Vrms
E1-E8	Distortion	IV	X	0.0	0.2	0.0	0.2	0.0	0.2	0.0	0.2	%
F	Resistance	V	X	4,465	4,935	4,465	6,345	4,230	5,170	4,230	6,580	Ω
G1, G2	Power supply transient rise time	VI	S	1.1	2.1	1.0	2.1	1.0	2.1	1.0	2.1	ms
H1-H8	Input current, gate "on"	VII	S	0.5	60	0.5	60	0.5	80	0.5	80	µA
J1-J8	Input current, gate "off"	VII	S	0.0	0.1	0.0	0.1	0.0	1.2	0.0	1.2	µA
K1-K16	Analog switch "on" resistance (see note 4)	VIII	X	0.01 (0.01)	10 (10)	0.01 (0.01)	10 (10)	0.01 (0.01)	13 (13)	0.01 (0.01)	15 (15)	Ω mV
L1	Power supply current, gate "on", plus supply current	IX	S	4	60	4	60	4	65	4	65	mA
L2	Power supply current, gate "on", minus supply current	IX	S	2	30	2	30	2	35	3	35	mA
L3	Power supply current, gate "on", reference supply current	IX	S	2	40	2	40	2	45	2	45	mA
L4-L6	Power supply current, gate "off", plus, minus, and reference supply current	IX	S	0.0	100	0.0	100	0.0	800	0.0	800	µA

See notes at end of table.

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TABLE I. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M8, M17-M24	Turn on time	X	S	0	1.0	0	1.0	0	1.5	0	1.5	µs
M9-M16, M25-M32	Turn off time	X	S	0	2.5	0	2.5	0	3.7	0	3.7	µs
N1-N16	Crosstalk (see note 4)	XI	S	0 (0)	2.83 (40)	0 (0)	2.83 (40)	0 (0)	2.83 (40)	0 (0)	2.83 (40)	mVpp/Vpp mVp-p
P1-P16	Source and drain leakage check	XII	X	0	8	0	8	0	8	0	8	mV

NOTES:

1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.
4. Test conditions K and N have two sets of limits given. The top set are the limits of the module in appropriate units. The bottom set in parenthesis are the equivalent limits resulting from the test method specified. The applicable table gives the formulas used in converting from the measured values to the appropriate units.

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6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

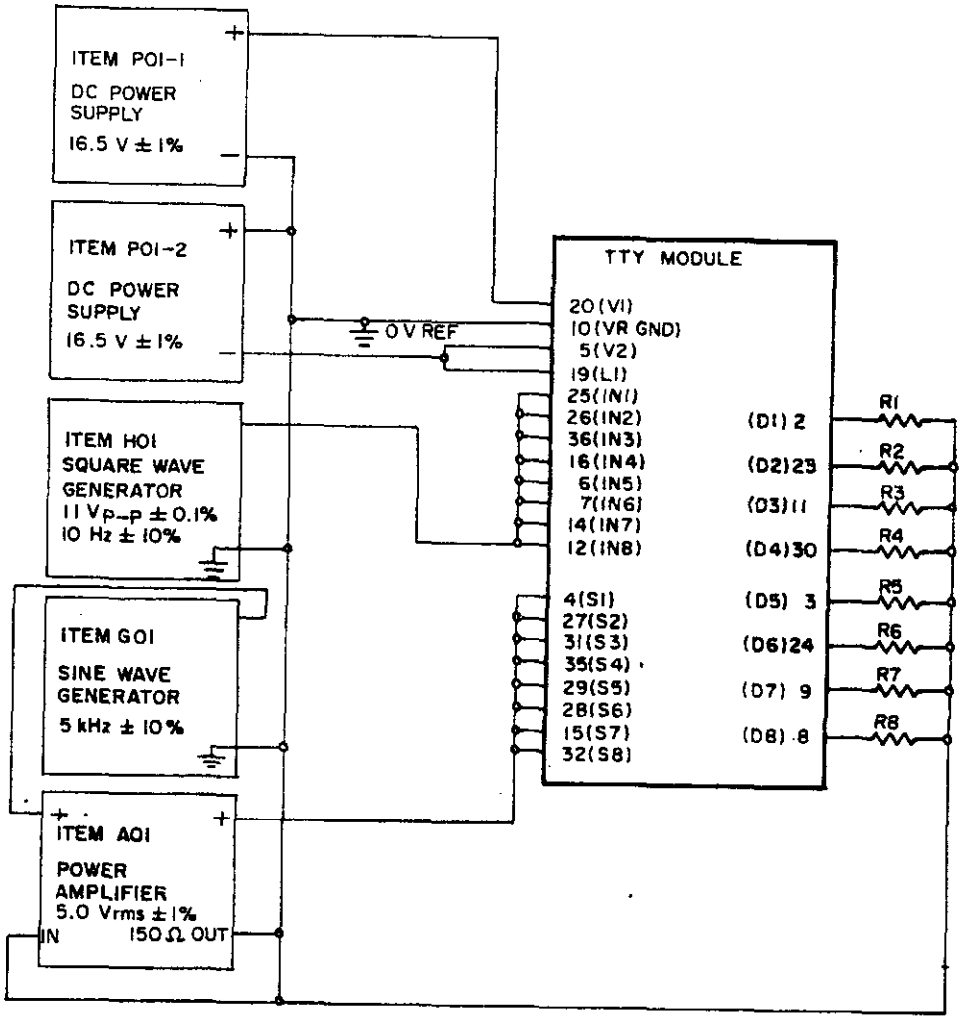
6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life.

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project MISC-NC71-14)

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NOTES:
1. R1-R8 = 235Ω ±1%, 1/2 W.
2. Items AOI, GOI, HOI, and R1-R8 are not connected during the vibration test.

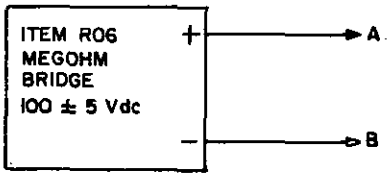
FIGURE 2. Operating temperature, transient thermal, vibration, and life test circuit.

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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	All pins	Frame	10	∞	MΩ	3
A2	Isolation	2,3,4,5,6,7,8, 9,10,11,12,14, 15,16,19,20,23, 24,25,26,27,28, 29,30,31,32,35, 36	1,13,17,18, 21,22,33,34, 37,38,39,40	10	∞	MΩ	3

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TTY MODULE	
20(VI)	
10(VR GND)	
5(V2)	
19(LI)	
25(IN1)	
26(IN2)	(D1) 2
36(IN3)	
16(IN4)	(D2)23
6(IN5)	
7(IN6)	(D3)11
14(IN7)	
12(IN8)	(D4)30
4(S1)	(D5) 3
27(S2)	
31(S3)	(D6)24
35(S4)	
29(S5)	(D7) 9
28(S6)	
15(S7)	(D8) 8
32(S8)	

WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

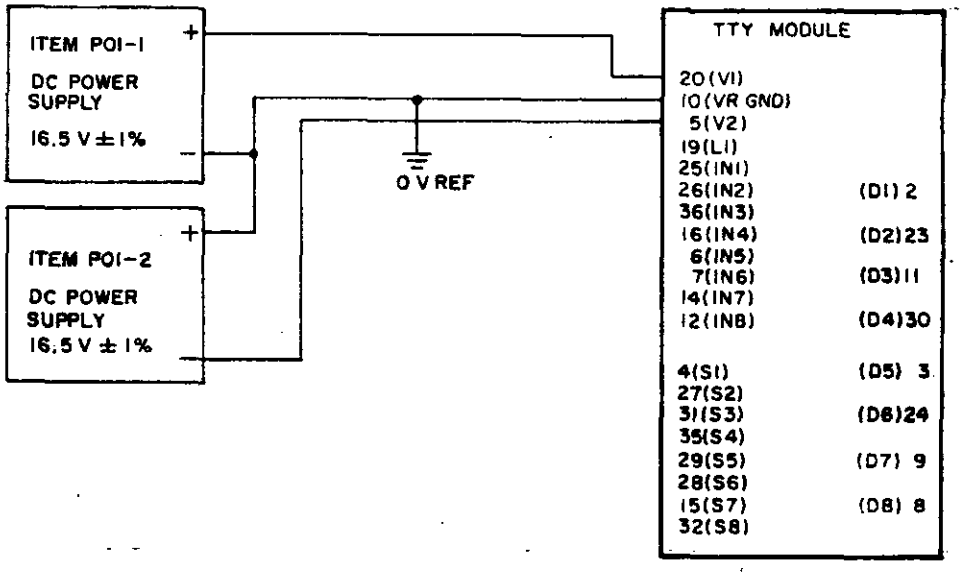
FIGURE 3. *Isolation test circuit for table II.*

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TABLE III. Power supply sequence test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
B	Power supply sequence	---	---	---	4

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NOTE: Turn on items P01-1 and P01-2 in the following sequence:

a. Turn on item P01-1 first and then item P01-2.
b. *Turn off items P01-1 and P01-2.*
c. Turn on item P01-2 first and then item P01-1.

FIGURE 4. Power supply sequence test circuit for table III.

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TABLE IV. Analog switch feed thru and distortion test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Analog switch feed thru, switch "off"	4	2	0.0	12.5	mVrms	5
C2		27	23				
C3		31	11				
C4		35	30				
C5		29	3				
C6		28	24				
C7		15	9				
C8		32	8				
C9		2	4				
C10		23	27				
C11		11	31				
C12		30	35				
C13		3	29				
C14		24	28				
C15		9	15				
C16	Analog switch feed thru, switch "off"	8	32	0.0	12.5	mVrms	
D1	Analog switch feed thru, switch "on"	4	2	4.9	5.1	Vrms	5

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TABLE IV. Analog switch feed thru and distortion test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D2	Analog switch feed thru, switch "on"	27	23	4.9	5.1	Vrms	5
D3		31	11				
D4		35	30				
D5		29	3				
D6		28	24				
D7		15	9				
D8		32	8				
D9		2	4				
D10		23	27				
D11		11	31				
D12		30	35				
D13		3	29				
D14		24	28				
D15		9	15				
D16	Analog switch feed thru, switch "on"	8	32	4.9	5.1	Vrms	5

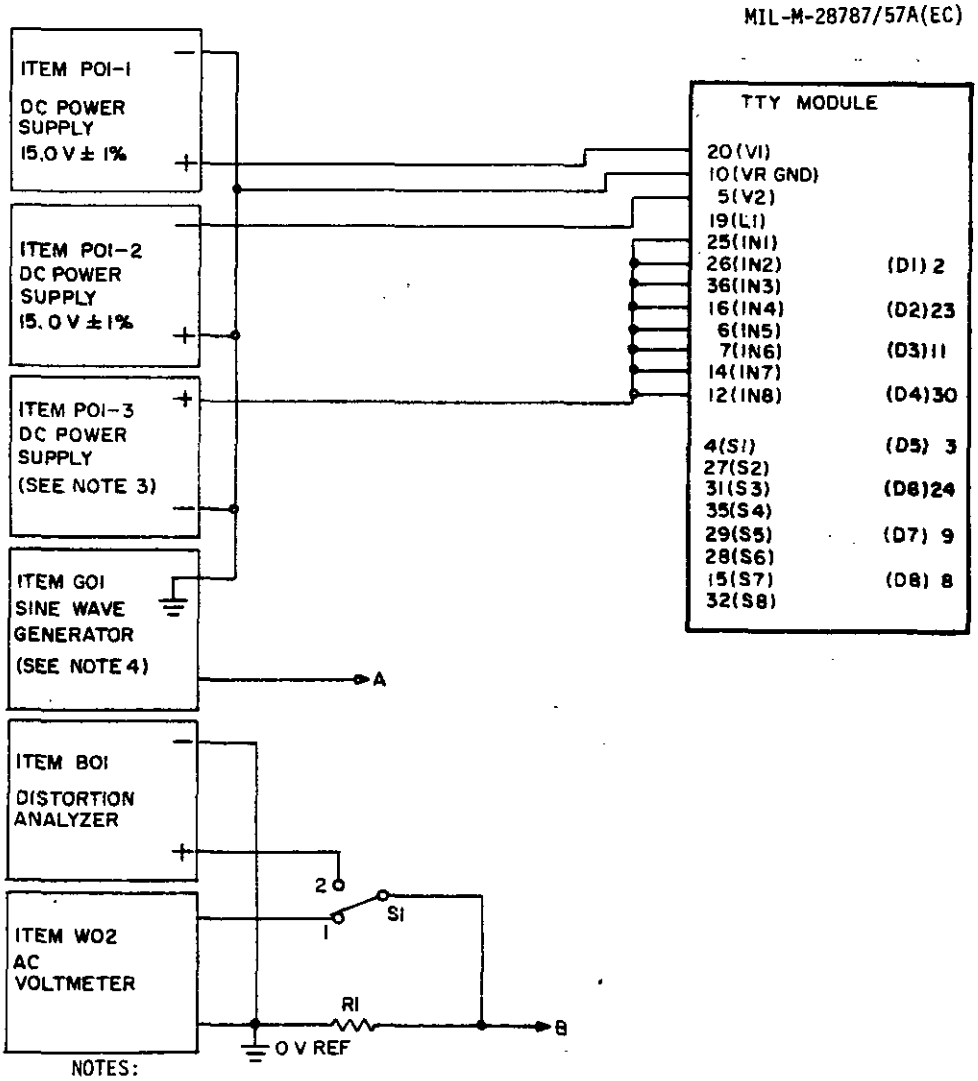
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TABLE IV. Analog switch feed thru and distortion test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Distortion (see note)	4	2	0.0	0.2	%	5
E2		27	23				
E3		31	11				
E4		35	30				
E5		29	3				
E6		28	24				
E7		15	9				
E8	Distortion (see note)	32	8	0.0	0.2	%	5

NOTE: Distortion introduced by module = Measured distortion output minus measured distortion input.

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NOTES:
1. RI = 4.7 kΩ ±1%, 1/10 W.
2. SI = SPDT switch.
3. Set item POI-3 as follows:

   a. Test conditions C1-C16 at 0.80 ±0.01 V.
   b. Test conditions D1-D16 at 2.41 ±0.01 V.
   c. Test conditions E1-E8 at 2.41 ±0.01 V.

4. Set item GOI as follows:

   a. Test conditions C1-C16 at 5.0 Vrms ±1%, 10 kHz ±1%.
   b. Test conditions D1-D16 at 5.0 Vrms ±1%, 10 kHz ±1%.
   c. Test conditions E1-E8 at 4.0 Vrms ±2%, 10 kHz ±1%.

5. Set switch SI as follows:

   a. Test conditions C1-C16 at position 1.
   b. Test conditions D1-D16 at position 1.
   c. Test conditions E1-E8 at position 2.

FIGURE 5. Analog switch feed thru and distortion test circuit for table IV.

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TABLE V. Resistance test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
F	Resistance	4,465	4,935	Ω	6

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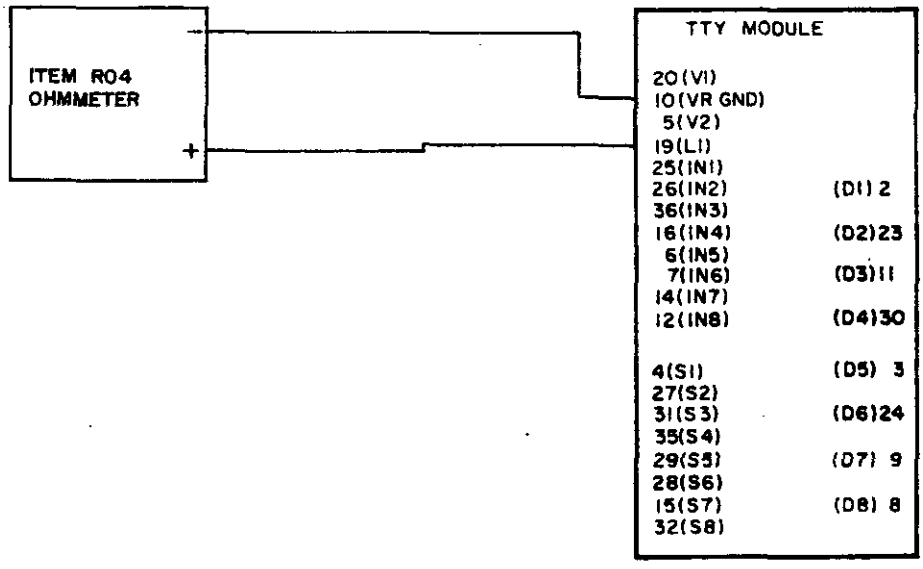


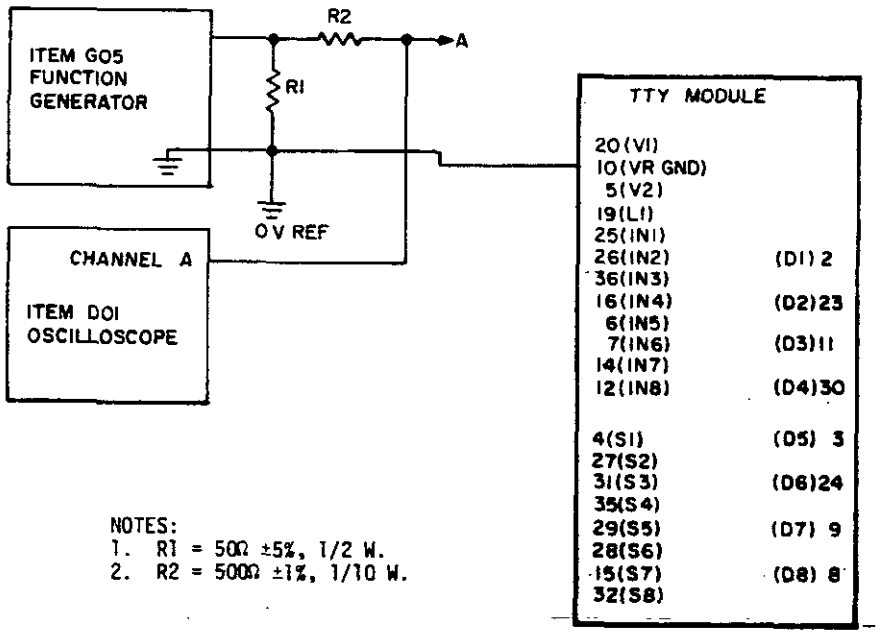
FIGURE 6. Resistance test circuit for table V.

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TABLE VI. Power supply transient rise time test.

Test condition	Parameter	Connect	Initial limits 25° +5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
G1	Power supply transient rise time	20	1.1	2.1	ms	7,8
G2	Power supply transient rise time	5	1.1	2.1	ms	7,8

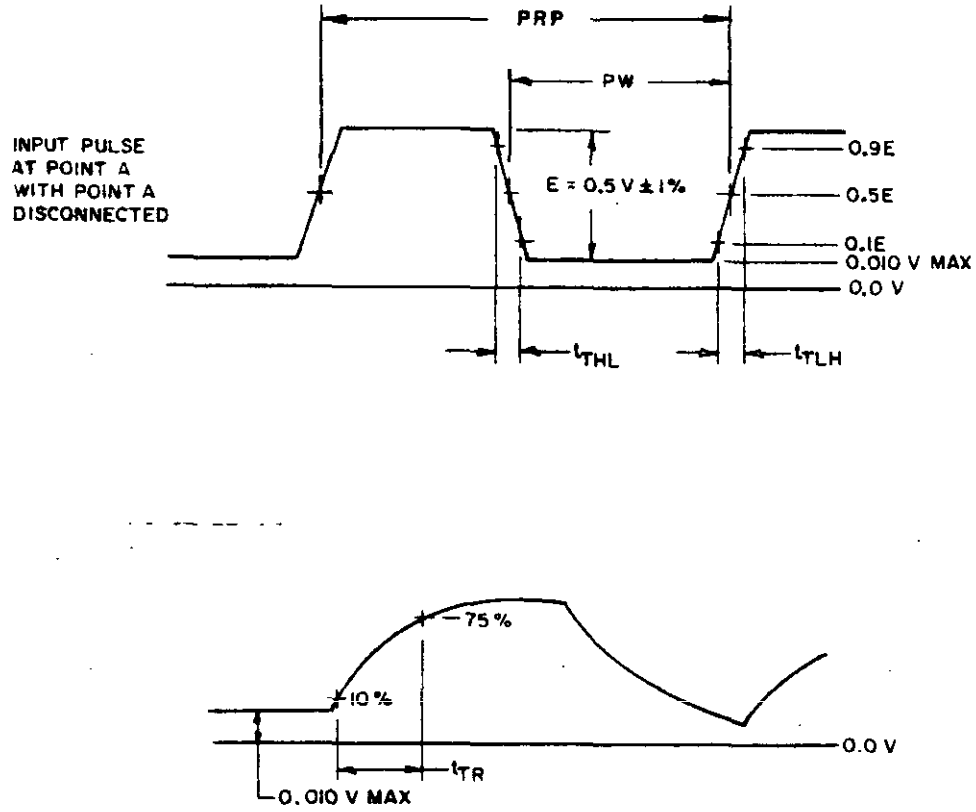
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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. R2 = 500Ω ±1%, 1/10 W.

FIGURE 7. *Power supply transient rise time test circuit for table VI.*

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NOTES:
1. Input pulse parameters:
   PW = 10 ms ±10%.
   PRP = 20 ms ±10%.
   $t_{TLH} = t_{THL}$ < 100 ns.
2. Measure the transient rise time (t_{TR}) between the 10% and 75% points of the pulse.

FIGURE 8. Power supply transient rise time test circuit waveforms for table VI.

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TABLE VII. Input current test.

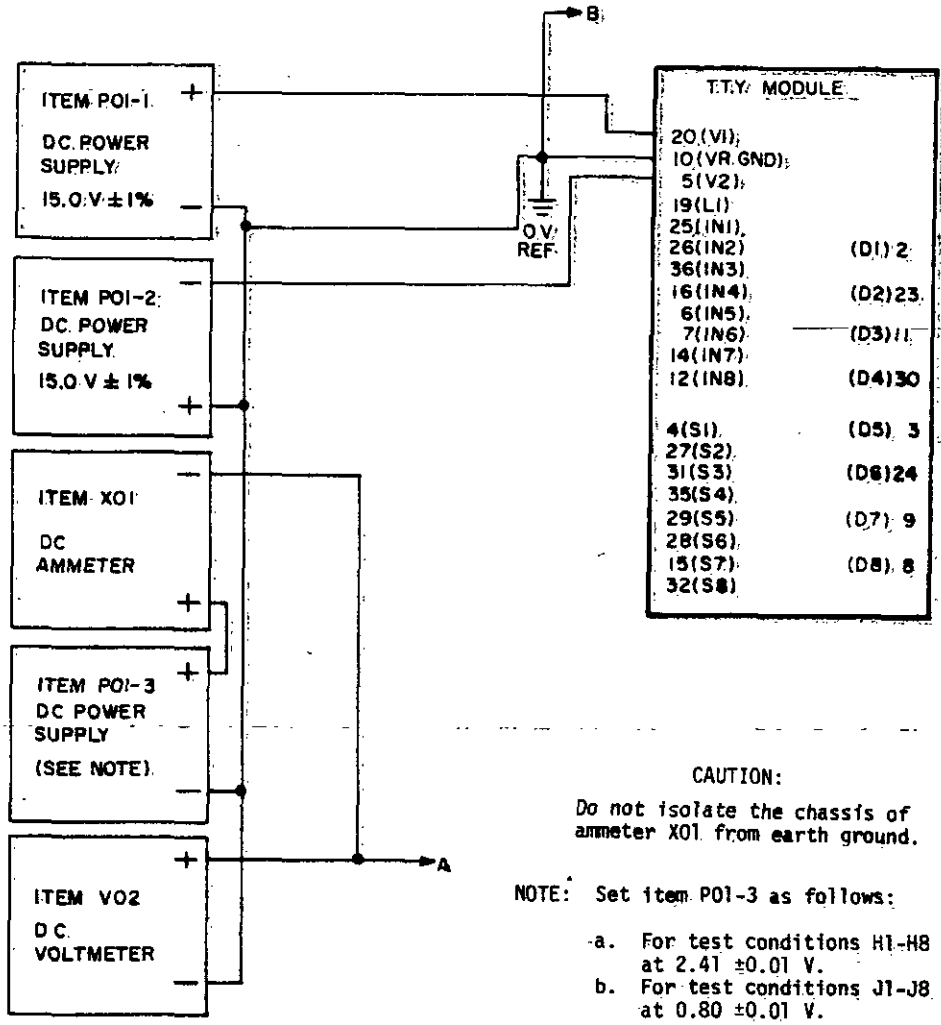
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	Input current, gate "on"	25	6,7,12,14,16, 26,36	0.5	60	µA	9
H2		26	6,7,12,14,16, 25,36				
H3		36	6,7,12,14,16, 25,26				
H4		16	6,7,12,14,25, 26,36				
H5		6	7,12,14,16,25, 26,36				
H6		7	6,12,14,16,25, 26,36				
H7		14	6,7,12,16,25, 26,36				
H8	Input current, gate "on"	12	6,7,14,16,25, 26,36	0.5	60		
J1	Input current, gate "off"	25	6,7,12,14,16, 26,36	0.0	0.1		
J2		26	6,7,12,14,16, 25,36				
J3		36	6,7,12,14,16, 26,26				
J4	Input current, gate "off"	16	6,7,12,14,25, 26,36	0.0	0.1	µA	9

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TABLE VII. Input current test -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J5	Input current, gate "off"	6	7,12,14,16,25, 26,36	0.0	0.1	µA	9
J6	↓	7	6,12,14,16,25, 26,36	↓	↓	↓	↓
J7	↓	14	6,7,12,16,25, 26,36	↓	↓	↓	↓
J8	Input current, gate "off"	12	6,7,14,16,25, 26,36	0.0	0.1	µA	9

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTE: Set item PO1-3 as follows:

a. For test conditions H1-H8 at 2.41 ±0.01 V.
b. For test conditions J1-J8 at 0.80 ±0.01 V.

FIGURE 9. Input current test circuit for table VII.

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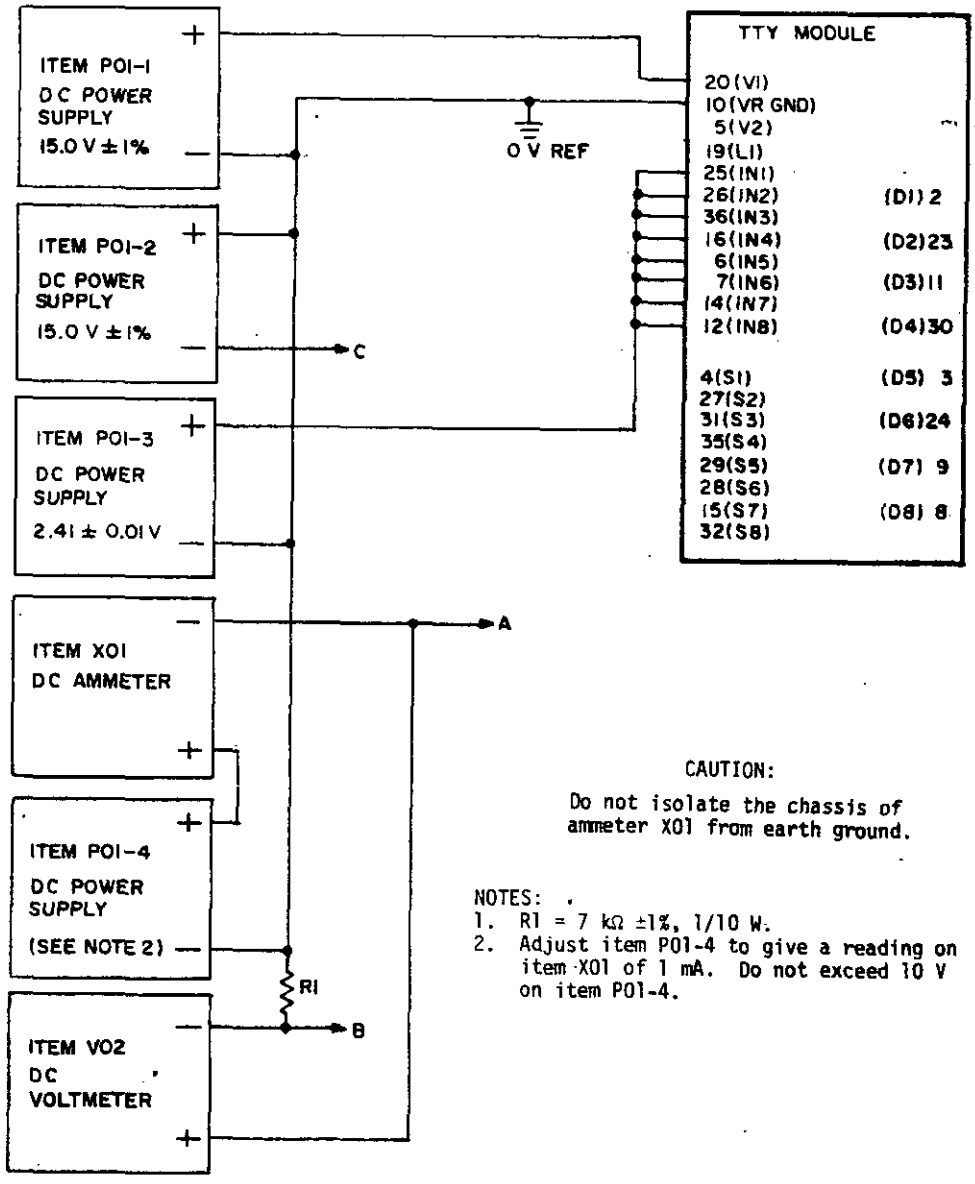
TABLE VIII. Analog switch "on" resistance test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	Analog switch "on" resistance, V1 and V2 applied (see note)	4	2	5	0.01	10	mV	10
K2		31	11					
K3		29	3					
K4		15	9					
K5		27	23					
K6		35	30					
K7		28	24					
K8	Analog switch "on" resistance, V1 and V2 applied (see note)	32	8	5				
K9	Analog switch "on" resistance, V1 applied (see note)	4	2	Open				
K10		31	11					
K11		29	3					
K12		15	9					
K13		27	23					
K14		35	30					
K15		28	24					
K16	Analog switch "on" resistance, V1 applied (see note)	32	8	Open	0.01	10	mV	10

NOTE: The analog switch "on" resistance in ohms may be calculated as follows:

$$R_{on} = \frac{\text{Item V02 reading}}{\text{Item X01 reading}}$$

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 7 kΩ ±1%, 1/10 W.
2. Adjust item PO1-4 to give a reading on item XO1 of 1 mA. Do not exceed 10 V on item PO1-4.

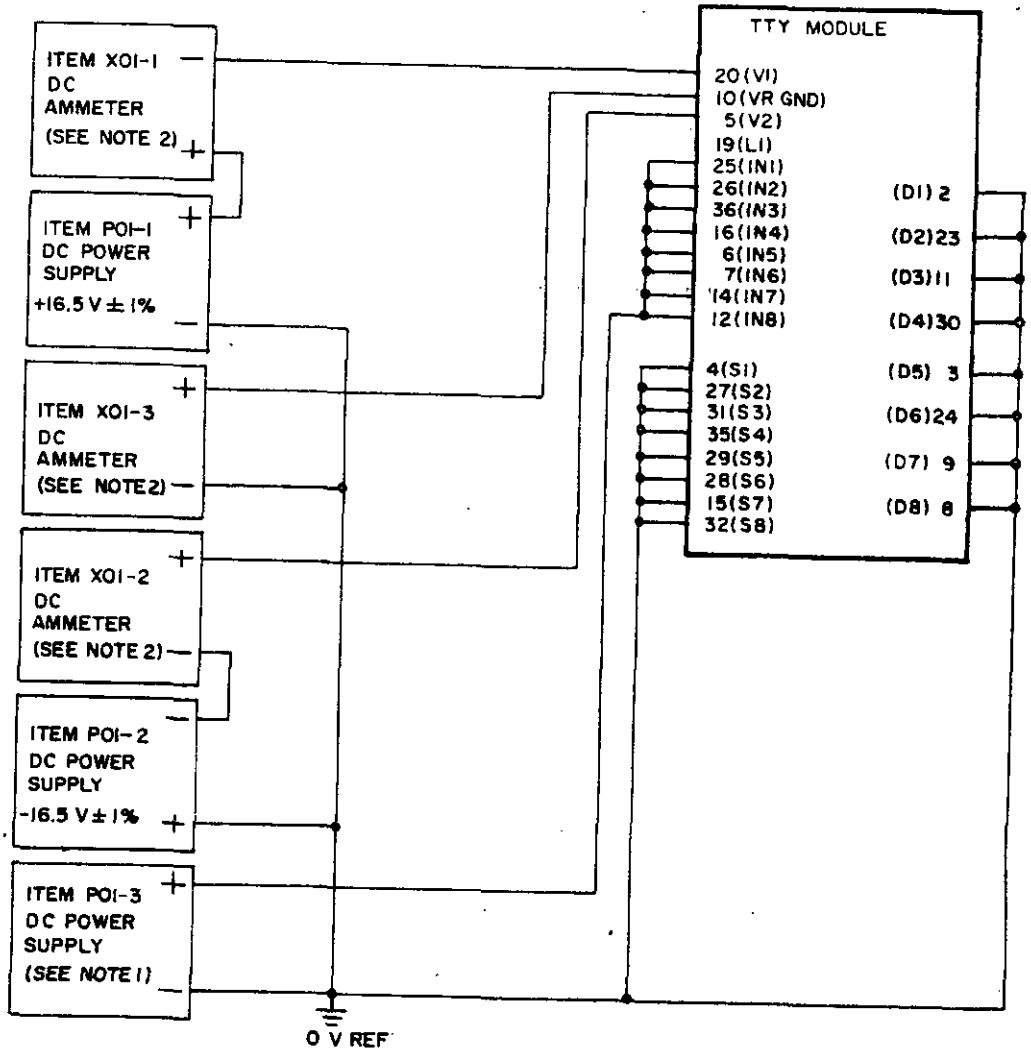
FIGURE 10. Analog switch "on" resistance test circuit for table VIII.

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TABLE IX. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
L1	Power supply current, gate "on", plus supply current	4	60	mA	11
L2	Power supply current, gate "on", minus supply current	2	30	mA	
L3	Power supply current, gate "on", reference supply current	2	40	mA	
L4	Power supply current, gate "off", plus supply current	0.0	100	µA	
L5	Power supply current, gate "off", minus supply current	0.0	100	µA	
L6	Power supply current, gate "off", reference supply current	0.0	100	µA	11

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:

1. Set item POI-3 as follows:

   a. For test conditions L1-L3 at 5.5 V ±1%.
   b. For test conditions L4-L6 at 0.8 V ±1%.

2. Read outputs as follows:

   a. Test conditions L1 and L4 on item XOI-1.
   b. Test conditions L2 and L5 on item XOI-2.
   c. Test conditions L3 and L6 on item XOI-3.

FIGURE 11. Power supply current test circuit for table IX.

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TABLE X. Turn on and turn off time test.

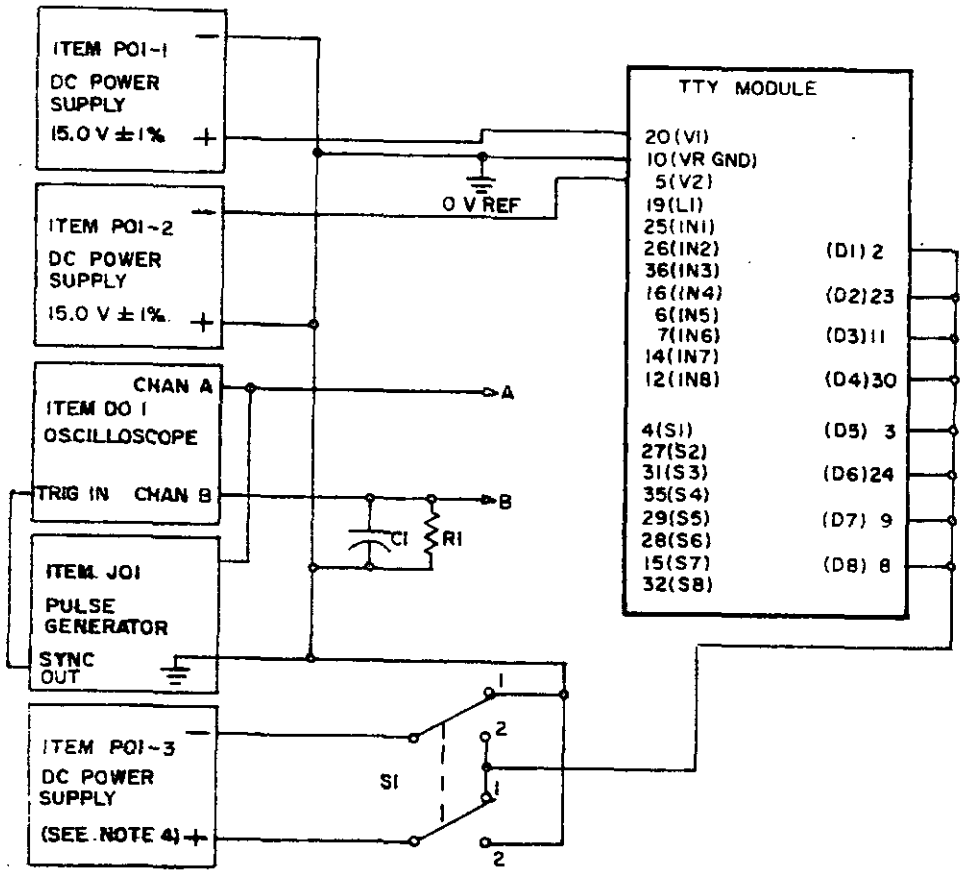
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Turn on time, plus swing	25	4	0	1.0	µs	12,13
M2		36	31				
M3		6	29				
M4		14	15				
M5		26	27				
M6		16	35				
M7		7	28				
M8	Turn on time, plus swing	12	32		1.0		
M9	Turn off time, plus swing	25	4		2.5		
M10		36	31				
M11		6	29				
M12		14	15				
M13		26	27				
M14		16	35				
M15		7	28				
M16	Turn off time, plus swing	12	32	0	2.5	µs	12,13

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TABLE X. *Turn on and turn off time test* -Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M17	Turn on time, minus swing	25	4	0	1.0	µs	12,13
M18		36	31				
M19		6	29				
M20		14	15				
M21		26	27				
M22		16	35				
M23		7	28				
M24	Turn on time, minus swing	12	32		1.0		
M25	Turn off time, minus swing	25	4		2.5		
M26		36	31				
M27		6	29				
M28		14	15				
M29		26	27				
M30		16	35				
M31		7	28				
M32	Turn off time, minus swing	12	32	0	2.5	µs	12, 13

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NOTES:
1. R1 = 1 kΩ ±1%, 1/10 W.
2. C1 = 35 pF ±10%, 20 Vdc.
3. S1 = DPDT switch.
4. Set item P01-3 as follows:

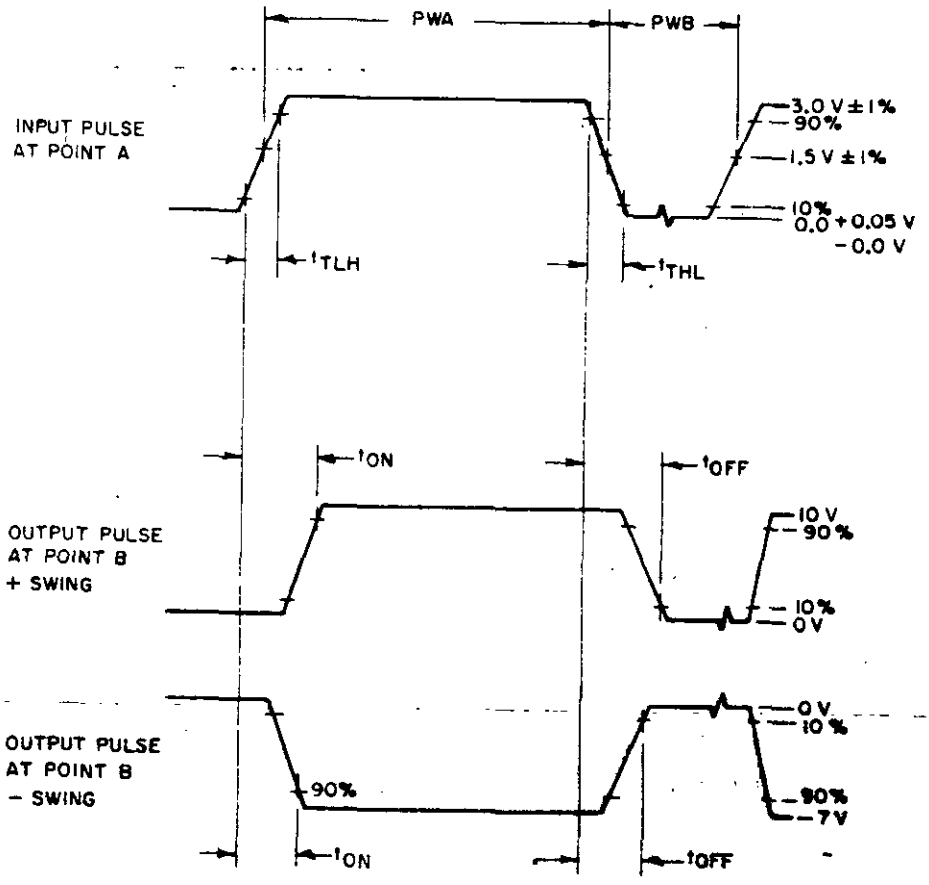
   a. For test conditions M1-M16 at +10.0 V ±1%.
   b. For test conditions M17-M32 at -7.0 V ±1%.

5. Set switch S1 as follows:

   a. For test conditions M1-M16 at position 1.
   b. For test conditions M17-M32 at position 2.

FIGURE 12. *Turn on and turn off time test circuit for table X.*

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NOTE: Input pulse parameters:
PWA = 5 µs ±1%.
PWB = 45 µs ±10%.
$t_{TLH} = t_{THL} \leq 50$ ns.

FIGURE 13. Turn on and turn off time test circuit waveforms for table X.

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TABLE XI. Crosstalk test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	Crosstalk (see note)	25	6,7,12, 14,16, 26,36	15,27, 28,29, 31,32, 35	2	0	40	mV_{p-p}	14
N2		25	6,7,12, 14,16, 26,36	15,27, 28,29, 31,32, 35	2				
N3		26	6,7,12, 14,16, 25,36	4,15, 28,29, 31,32, 35	23				
N4		26	6,7,12, 14,16, 25,36	4,15,28 29,31, 32,35	23				
N5		36	6,7,12, 14,16, 25,26	4,15,27 28,29, 32,35	11				
N6		36	6,7,12, 14,16, 25,26	4,15, 27,28, 29,32, 35	11				
N7		16	6,7,12, 14,25, 26,36	4,15, 27,28, 29,31, 32	30				
N8	Crosstalk (see note)	16	6,7,12, 14,25, 26,36	4,15,27 28,29, 31,32	30	0	40	mV_{p-p}	14

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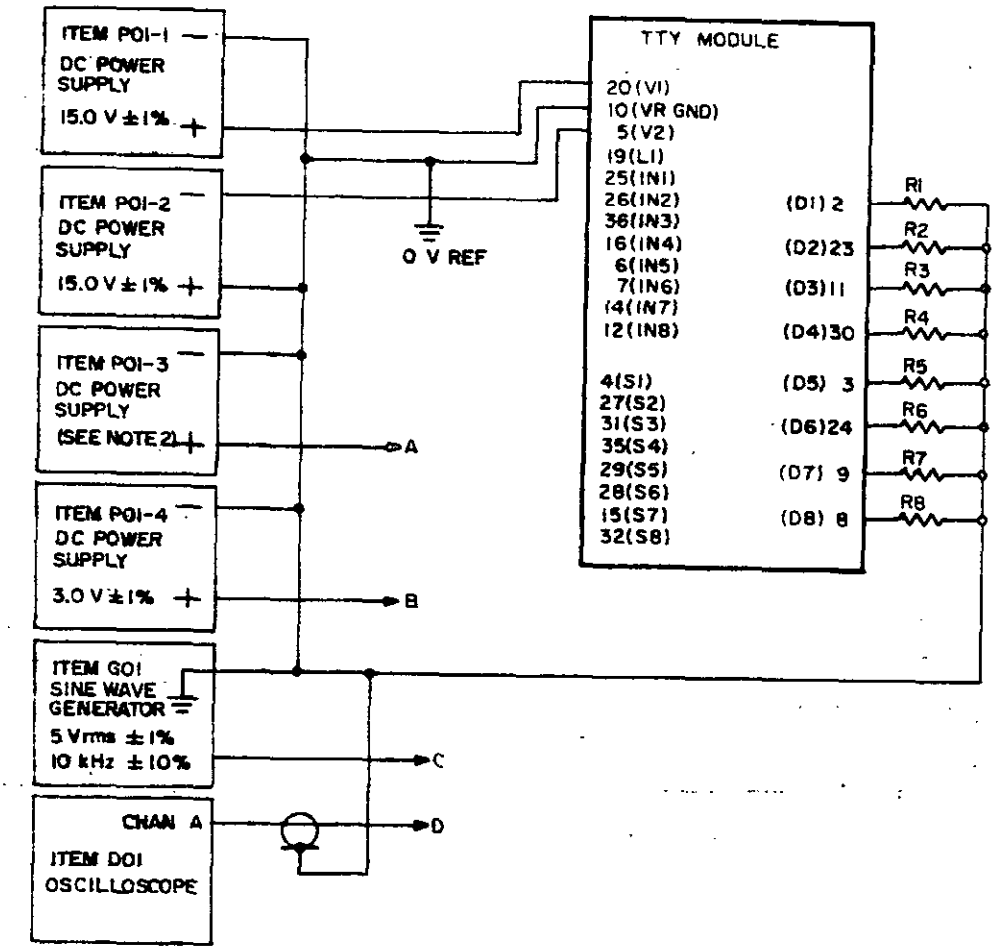
TABLE XI. Crosstalk test -Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N9	Crosstalk (see note)	6	7,12, 14,16, 25,26, 36	4,15, 27,28, 31,32, 35	3	0	40	mV_{p-p}	14
N10		6	7,12, 14,16, 25,26,36	4,15, 27,28, 31,32, 35	3				
N11		7	6,12, 14,16, 25,26, 36	4,15, 27,29, 31,32, 35	24				
N12		7	6,12, 14,16, 25,26, 36	4,15, 27,29, 31,32, 35	24				
N13		14	6,7,12, 16,25, 26,36	4,27, 28,29, 31,32, 35	9				
N14		14	6,7,12, 16,25, 26,36	4,27, 28,29, 31,32, 35	9				
N15		12	6,7,14, 16,25, 26,36	4,15, 27,28, 29,31, 35	8				
N16	Crosstalk (see note)	12	6,7, 14,16, 25,26, 36	4,15, 27,28, 29,31, 35	8	0	40	mV_{p-p}	14

NOTE: Crosstalk may be expressed in mV_{p-p}/V_{p-p} as follows:

$$\text{Crosstalk } (mV_{p-p}/V_{p-p}) = \frac{mV_{p-p} \text{ at point D/14.14}}{1\ V_{p-p}}$$

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NOTES:
1. R1-R8 = 4.7 kΩ ±1%, 1/10 W.
2. Set item P01-3 as follows:

   a. For test conditions N1, N3, N5, N7, N9, N11, N13, and N15 at 2.4 V ±1%.
   b. For test conditions N2, N4, N6, N8, N10, N12, N14, and N16 at 0.8 V ±1%.

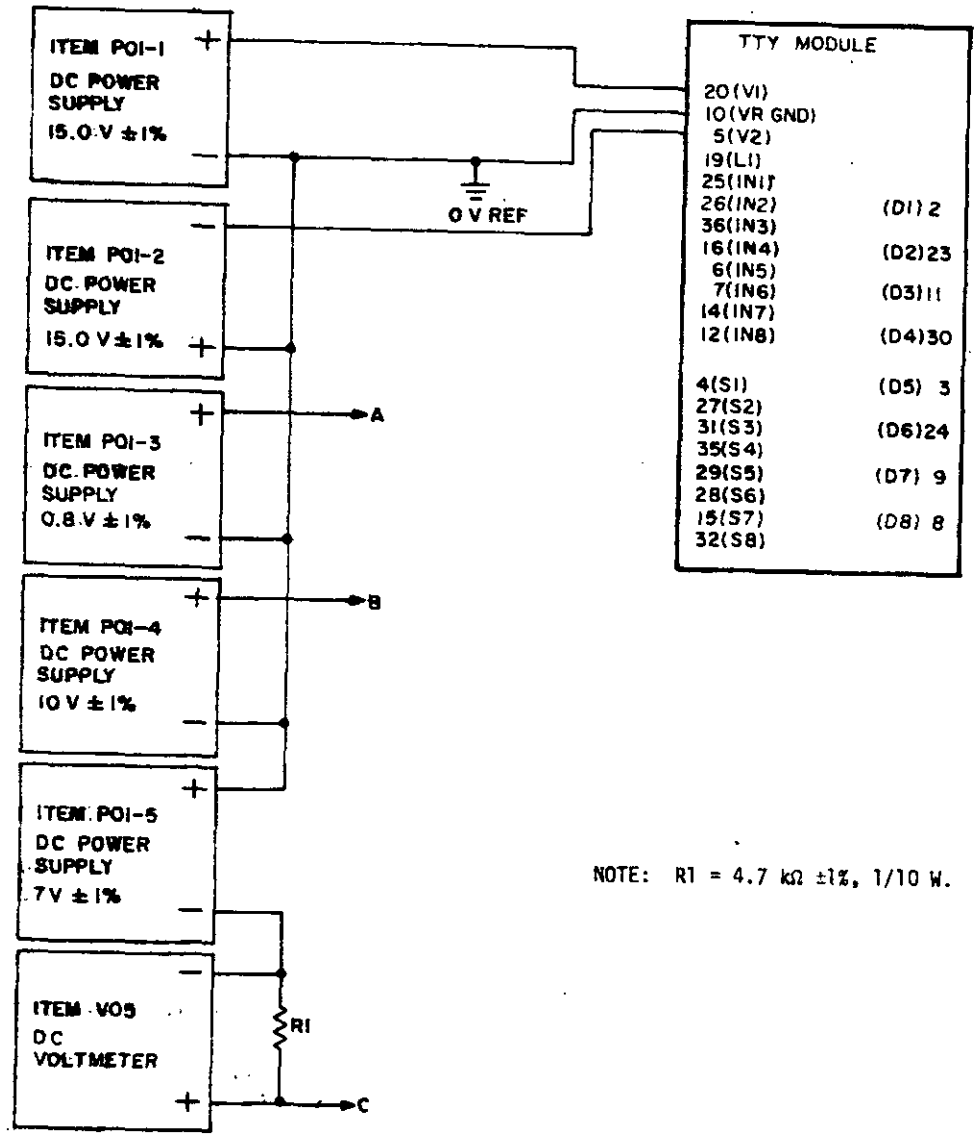
FIGURE 14. Crosstalk test circuit for table XI.

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TABLE XII. Source and drain leakage check.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	Source leakage check	25	4	2	0	8	mV	15
P2		26	27	23				
P3		36	31	11				
P4		16	35	30				
P5		6	29	3				
P6		7	28	24				
P7		14	15	9				
P8	Source leakage check	12	32	8				
P9	Drain leakage check	25	2	4				
P10		26	23	27				
P11		36	11	31				
P12		16	30	35				
P13		6	3	29				
P14		7	24	28				
P15		14	9	15				
P16	Drain leakage check	12	8	32	0	8	mV	15

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NOTE: R1 = 4.7 kΩ ±1%, 1/10 W.

FIGURE 15. Source and drain leakage check test circuit for table XII.

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